



M-Tech Production & Marketing

Technology | Training | Marketing | Projects

TASK 7 REPORT

Internship Program — Batch 2026

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1. Introduction

This report presents the work completed for **Task 07** of the M-Tech Production & Marketing AI/ML Internship Program, Batch 2026. The internship focuses on practical exposure to core concepts in Artificial Intelligence and Machine Learning, equipping interns with skills applicable to real-world technology and marketing challenges.

M-Tech Production & Marketing is a forward-looking company based in Haripur, KPK, specializing in Technology, Training, Marketing, and Projects. The internship is structured in a 9-Phase roadmap designed to progressively build theoretical knowledge and hands-on project experience.

2. Objectives

- To understand the concept of Ensemble Learning.
- To learn how combining multiple models improves prediction accuracy.
- To study the working principles of Random Forest.
- To understand the fundamentals of XGBoost and LightGBM.
- To compare different ensemble algorithms based on performance and efficiency.
- To explore practical applications of ensemble methods in AI and Machine Learning.

3. Task Description

In this experiment, I studied Ensemble Learning, a machine learning technique that combines multiple models to produce more accurate and reliable predictions than a single model. I explored Random Forest, which builds multiple decision trees and combines their predictions to improve accuracy and reduce overfitting.

I also learned about XGBoost (Extreme Gradient Boosting) and LightGBM (Light Gradient Boosting Machine), two advanced boosting algorithms designed for high performance, faster training, and better predictive accuracy. The experiment also

covered the advantages, differences, and real-world applications of these ensemble methods.

4. Work Done

During this experiment, I completed the following activities:

- Studied the fundamentals of Ensemble Learning.
- Learned the difference between Bagging and Boosting techniques.
- Explored the working mechanism of Random Forest.
- Understood how Random Forest combines multiple decision trees to improve prediction accuracy.
- Studied the concepts of XGBoost, including gradient boosting and sequential learning.
- Learned how LightGBM improves training speed and efficiently handles large datasets.
- Compared Random Forest, XGBoost, and LightGBM based on performance, speed, accuracy, and scalability.
- Explored real-world applications of ensemble methods in fraud detection, customer churn prediction, healthcare, finance, recommendation systems, and predictive analytics.

5. Challenges Faced

- Understanding the difference between Bagging and Boosting algorithms.
- Learning how multiple decision trees work together to improve predictions.
- Differentiating between XGBoost and LightGBM in terms of performance and efficiency.
- Understanding when to select a specific ensemble method for different machine learning tasks.

6. Key Learnings

Through this experiment, I learned:

- The importance of Ensemble Learning in improving machine learning model performance.
- How Random Forest reduces overfitting by averaging predictions from multiple decision trees.
- How XGBoost improves prediction accuracy through gradient boosting.
- How LightGBM provides faster training and efficient processing for large datasets.
- The advantages and limitations of different ensemble methods.
- Practical applications of ensemble learning in solving complex real-world machine learning problems.

7. Conclusion

This experiment provided a comprehensive understanding of Ensemble Methods, one of the most powerful techniques in modern machine learning. I learned how Random Forest, XGBoost, and LightGBM improve prediction accuracy by combining multiple models and reducing model errors. These algorithms are widely used in industry due to their high performance, scalability, and reliability.

The knowledge gained from this experiment has strengthened my understanding of advanced machine learning techniques and will support the development of more accurate and robust AI solutions in future projects.

Intern Signature: Hamza Badshah **Supervisor:** _____

Name: Hamza Badshah

Name:

Date: 12 July 2026

Designation: